

LM1 ETF

Mechanics

open-end fund: 份额变化, AP 购入 + closed-end fund 特质 (二级市场交易)

一级市场交易: in-kind 实物申购赎回, (一篮子股票) 买入 Create, 赎回 Redeem

二级市场交易: Intraday 日内交易, price $\approx$ NAV

AP 同时在一级、二级市场交易, 套利

一: 股市值 = ETF 价值 NAV
二: ETF 由 O/S 决定

NAV
PETF

Premium 时 - 一级 Create
Discount 时 - 一级 Redeem

AP $\xleftrightarrow[\text{份额}]{\text{Stock}}$ 基金公司

Arbitrage Gap 与 Trading cost 同向

Costs & Risks

1. Trading Costs

Round-trip trading cost (%) = (one way commission + 0.5 $\times$ bid-ask spread) $\times$ 2.

(一来一回, 单方向成本 $\times$ 2) = two way commission + bid-ask spread

Holding period cost (%) = Management fee + Round-trip trading cost.

与时间有关 $\times \frac{1}{2}$

与时间无关

Bid-Ask Spread: 买 Bid $\rightarrow$ AP 作为 Dealer / Market Maker $\rightarrow$ 卖 Ask

① Creation/Redemption Fee

② 1 股 B-A Spread

③ Risk Compensation

① 影响因素

Liquidity of ETF
Liquidity of underlying asset
Market Maker 风险越小, spread 越小.

流动性越好, spread 越小.

② 主要素

spread $\leq$ 主要素的 Combination, 是 spread 上限

③ Premium & Discount to NAV

计算 End-of-day % = (ETF Price - NAV) / NAV 日交易结束后公布的 NAV

Intraday % = (ETF Price - iNAV) / iNAV indicated NAV, 计算 Fair Value

原因: Time Difference (跨时区), State Pricing (ETF 与 Underlying Security 交易流动性差异)

2. Tracking Error 跟踪误差

年化的

针对 Index-tracking ETF, 与 fund one day or periodic 差距的标准差

Tracking Error = 0, 不代表完全是 passive investment.
Tracking Error 距 0 越远, 越偏离 passive investment.

ETF	HS 300
day1	$R_1 - R_1^I = d_1$
day2	$R_2 - R_2^I = d_2$
$\vdots$	$\vdots$
收益率差的标准差	$\sigma(d_i)$

$$T.E = \sqrt{\frac{\sum (R_i - R_i^I)^2}{n-1}}$$

原因: Fees and expense.

Sampling and optimization. (AP - 篮子股票中有替代品, 实物与 Index 有差距)

Depository Receipt 存托凭证 (流动性差的股票可能以存托凭证给基金公司)

Index change

Fund Accounting practice (Fund, Index valuation practice difference).

Regulatory and tax requirement.

Asset Manager operation.

(导致 Index 好)

Security lending, ETF manager 给市场融券从而收费 "要花钱"

Foreign dividend recapture. 收到分红再投资.

3. Expense Ratios 费用率

ETF charge **lower fees** than mutual fund: 被动, 不与个人投资者打交道.

4. Tax Issues

① ETP Tax Fairness.

Mutual Fund 赎回时, 还投资者钱, capital gain 须纳税, 投资者们共担, 不满足 fairness.

ETF 赎回 in kind, not a taxable event. 但 ETF 二级市场谁赚钱谁纳税.

② ETF Tax Efficiency. (Capital Gain 最高)

ETF 赎回时挑选建仓成本最低的股票还给 AP, 将税收负担转移给 AP. 基金公司抬高自身成本少纳税

成本高 → NAV 低 → 将成本转给投资者.



Cost Factor	Function of Holding Period	Explicit or Implicit	ETFs	Mutual Funds
Management Fee	✓	Explicit	✓ (less)	✓
Tracking Error	✓	Implicit	✓ (less)	✓ (index funds)
Commissions	×	Explicit	✓ (some free)	
Bid-Ask Spreads	×	Implicit	✓	
Premium / Discount to NAV	×	Implicit	✓	
Portfolio Turnover	✓	Implicit	✓ (less)	✓
Taxable G/L	✓	Explicit	✓ (less)	✓
Security Lending	✓	Implicit	✓ (more)	✓

5. Risks

Counterparty risk

Settlement risk: 引入衍生工具后的结算风险.

Security lending

Fund closures: 倒闭, Soft closures 关停、定转.

Investor related risk

ETF in PM

Efficiency { Liquidity 过度产品.
Rebalance 构建. 挂钩特定 Index, 提供敞口.
基金经理空窗期.

Asset Exposure.
Active 管理.

★ LM2 Multifactor Model (7X)

Arbitrage Pricing Model (APT 套利定价模型)

1. 辨析: $ECR_p = R_f + \beta_{p,1} \times \lambda_1 + \beta_{p,2} \times \lambda_2 + \dots + \beta_{p,k} \times \lambda_k$
- 一系列系统性风险维度的线性组合.
 - CAPM 以 $ECR_m - R_f$ 代表系统性风险
 - APT 以 $\beta_{p,j} \times \lambda_j$ 共同解释系统性风险.
- $\beta_{p,j}$: sensitivity to factor j . 系数
 $\lambda_j = ECR_j - R_f$ expected risk premium for factor j .
 also call factor risk premium. 解释变量
 铁打的入, 流水的 β . 一套 APT 入不变, β 随 Portfolio 变动

2. Pure Factor Portfolio: factor j 的 $\beta = 1$, 其余 $\beta = 0$. 用于将抽象的东西实在化.

$$ECR_{\text{pure-factor-portfolio}} = R_f + 0 \cdot \lambda_1 + 0 \cdot \lambda_2 + \dots + 1 \cdot \lambda_j = R_f + \lambda_j = R_f + [ECR_j - R_f] = ECR_j$$

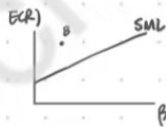
由 $R_f + \lambda_j$ 来倒推 Portfolio 线性组合.

3. APT Assumptions

- Factor model describes asset returns.
- Well-diversified, eliminate asset-specific asset.
- No arbitrage opportunity, markets are in equilibrium $\rightarrow ECR_p$.

Arbitrage Opportunity.

① CAPM 中:



Jensen's $\alpha_B = R_B - ECR_B > 0$ 实际 - 预期
 $= R_B - [R_f + \beta_B (ECR_m - R_f)]$ 根据 +, - 判断 long, short

long 1 份 B, short (1 - β_B) 份 R_f , short β_B 份 market portfolio.

② APT 中:

$$ECR_A = R_f + \beta_{A,1} \lambda_1 + \beta_{A,2} \lambda_2$$

$$ECR_B = R_f + \beta_{B,1} \lambda_1 + \beta_{B,2} \lambda_2$$

完全相同, 但实际 return 不同时, 低买高卖.
 return 高, return 低.

Multifactor Models. (7X)

1. Structures.

① Macroeconomic Factor Model.

$$R_i = ECR_i + \beta_{i,1} F_1 + \beta_{i,2} F_2 + \dots + \beta_{i,k} F_k + \varepsilon_i$$

$$R_i - ECR_i = \beta_{i,1} F_1 + \beta_{i,2} F_2 + \dots + \beta_{i,k} F_k + \varepsilon_i$$

回报率的 surprise = 一系列宏观因子 surprise 的线性组合.

Inflation, GDP growth, ...

采用时间序列数据, 先处理 F 再处理 β

F_k 是宏观因子 surprise = realized value - predicted value

铁打的 F, 流水的 β , ECR , ε .

ECR_i 可以通过 APT Model 得出.

算 β F 数值大小 Hs Factor 影响.

② Fundamental Factor Model

$$R_i = a_i + \beta_{i,1} F_1 + \beta_{i,2} F_2 + \dots + \beta_{i,k} F_k + \varepsilon_i$$

a_i 是回归出来的截距项.

F_k 是 cross-sectional difference (stock price, P/E, P/B, earning growth rate) 不是 surprise. 是 attribute

含义: 每一个标准差, R 增加多少

$$\beta_{i,k} \text{ 是 standardized beta. } \beta_{i,k} = \frac{\text{Value of attribute } k \text{ for asset } i - \text{Average value of attribute } k}{\sigma(\text{Values of attribute } k)}$$

该要素随行业平均好几个标准差

采用横截面数据, 先处理 β 再处理 F.

3 Statistical Factor Model

假定少, 经济意义难解释, 统计上显著就 ok.

Factor Analysis Model (F.A).
Principal Component Model (P.C.A).

⇒ 公共因子组合, 替代原因子
数据降维, 简单化, 但也丧失信息.

2. Application.

1 Return Attribution.

Active Return, 也称为 Excess Return = $R_p - R_{\text{benchmark}}$. 不同于 Jensen's $\alpha = R_p - \bar{R}(R_p)$!

Active Return = Factor Return (因子选择) + Security Selection Return (个股选择).

asset class selection

individual asset selection.

$\sum (R_j^p - R_j^B) \cdot F_j$
多配因子, 少配因子 → 超额收益.

$(R_p - R_B) + (E_p - E_B)$

2 Risk Attribution.

$$\text{Information Ratio} = \frac{\text{Active Return}}{\text{Active Risk}} = \frac{R_p - R_B}{\sigma(R_p - R_B)}$$

$$\text{Adjusted Sharp Ratio} = \frac{R_p - R_f}{\sigma_p}$$

Active Risk
= Tracking Error
= Relative Vol.

Active Risk Squared = Active Factor Risk (因子选择) + Active Specific Risk (个股股票 ϵ_i). (系统 + 非系统).
拆解方差 实际也是 squared.

3 Portfolio Construction and Decisions

Fama and French 基础上增加.

Carhart Four-Factor Model: $ECRP_t = R_f + \beta_{p,1} \times RMRF + \beta_{p,2} \times SMB + \beta_{p,3} \times HML + \beta_{p,4} \times WML$

$R_m - R_f$ Size Effect Value Effect Winner-Loser (惯性).

$R_p = \alpha_p + ECRP_t + \epsilon_i$ α 是 Jensen α , 代表超额

$R_p = \alpha_p + R_f + \beta_{p,1} \times RMRF + \beta_{p,2} \times SMB + \beta_{p,3} \times HML + \beta_{p,4} \times WML + \epsilon_i$

2. Measuring and Managing Market Risk.

Var.

1. Definition: 标准分位数, 最大损失, 最小损失, 取绝对值, 数越大风险越高.

若左偏 left-tail  则落在更左边, 风险更大.

% Var

$$\$Var = \% Var \times Up$$

2. Estimate Var 计算, 优缺点.

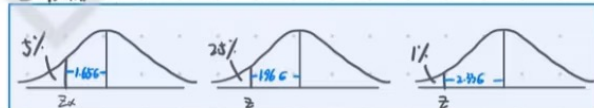
① Parametric Method. 参数法. 看清日 or 年.

假定收益率服从正态分布, $Var = |ECR| - Z_\alpha \times \sigma$

Square Root Rule: $Var(CT-day) = Var(one-day) \times \sqrt{T}$

方法缺点: $\left\{ \begin{array}{l} \text{参数不准则结果不准} \\ \text{收益结构可能不满足正态分布} \end{array} \right.$

3. 背诵



组合: $\mu_p = w_1 R_1 + w_2 R_2$
 $\sigma = \sqrt{w_1^2 R_1^2 + w_2^2 R_2^2 + 2w_1 w_2 R_1 R_2 \rho_{12}}$
 $Var_p = |\mu_p - Z_\alpha \cdot \sigma|$

② Historical Simulation Method 历史模拟法

从差到好排序 一刀切: 差(含 Var) | 好.

Var 5%:



直接 1% 得分位数

优点: 无须分布假设.

缺点: 认为历史数据可以代表未来.

③ Monte Carlo Simulation Method

模拟结果本身是概率分布, 得到分位数, Var 的本质

站在今天看未来

优点: 无须分布假设 (自己假设)

更 flexible, 未来路径建模随心所欲

缺点: 耗时耗力, 过于 flexible.

3. Extension

① Conditional Var (CVar).

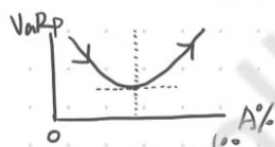
算术平均 Loss, 也称 Expected Shortfall. (ES).

② Incremental Var (IVAR).

管理规模扩大前后的 Var 之差.

③ Marginal Var (MVar).

Change in Var for \$1 or 1% 某资产规模扩大. (非整体管理规模).



MVar is the slope of Var-weighted curve for security A in portfolio.
 $MVar_A = 0$, 是 A 的最佳权重点.

④ Relative Var (RVar).

ex ante tracking error. 事前. 衡量偏离 benchmark

Relative Var
 $= \text{Tracking Error}$
 $= \text{Active Risk}.$

Other Risk Measures

1. Sensitivity Risk Measures

线性假定, y 对 single risk factor 微小变动的 respond. (其余变量保持不变). $\frac{\partial y}{\partial x}$ 求偏导

三个资产大类: ① Equity: 关注 β . (与 1 比较). $ECR_i = P_f + \beta_i \times [E(R_m) - P_f]$.

② Fixed Income: 关注 Duration & Convexity. $\Delta P/P = -\text{Duration} \times \Delta Y + \frac{1}{2} \times \text{Convexity} \times (\Delta Y)^2$.

③ Option: 关注 Delta, Gamma, Vega... $\Delta P_{\text{call}} = \text{Delta} \times \Delta S + \frac{1}{2} \times \text{Gamma} \times (\Delta S)^2 + \text{Vega} \times \Delta \sigma$

2. Scenario Risk Measures

Impact on Portfolio Value of significant changes in multiple risk factors. (single 也在内).

Approach: ① Historical scenario approach: 昨日重现.

② Hypothetical scenario approach: 历史未发生过.

Stress Test, extreme changes of risk factor, sustainability 持续性

Choices of Risk Measures.

4 机构角度:

① Banks

② Asset Managers | Traditional: Relative Return/Risk
Hedge Fund: Absolute Return/Risk

③ Pension Fund: AML!

④ Insurers | 财险 $\approx$ Banks
寿险 $\approx$ Pension Fund

Managing Market Risk

Risk Budgeting 风险预算: Determining overall risk appetite and allocate to sub-activities.

Capital Allocation 资本分配.

Position Limits 头寸限额.

Scenario Limits

Stop-loss Limits 止损

LM4 Back-testing & Simulation.

考点: 比较, 问题解决

Back-testing

① 定义: 用 historical data 去检验 Risk Trade-off 来检验好坏.

② Process: ① Strategy design: 时间不可太短 {
Downside Risk < VaR
最大回撤
Structural Change $\Rightarrow$ 前后分别 scenario analysis

② Historical Investment Simulation: Rolling Window (样本内模拟, 样本外检测)

③ Analysis of back-testing output: {
最大回撤: Max Drawdown
LT log 长期数据取对数

③ Problems: ① Look-ahead bias {
Reporting lag $\leftarrow$ Stable State.
Revisions

Index 重构: Survivorship bias $\leftarrow$ point-in-time data 是过去一节点最全的数据

② Data snooping (p-hacking) $\leftarrow$ {
Higher Hurdle
Rolling Window.

$\leftarrow$ 解决方法.

Simulation.

① H.S vs MCS

	Historical Simulation	Monte Carlo Simulation
优:	真实+易观.	$\vee$ what if / 观察灵活
缺:	$\times$ what if	主观+复杂.

② H.S. vs. Backtesting

{
H.S. 更随机. B.T. 用 historical data 更准确 (100%)
Sample Size < Simulation 次数 bootstrapping

③ M.C.S

① Model Calibration 模型校准 (用历史数据检测 M.C. 模型).

② 分布假设注意

- 1. 经济实质.
- 2. ρ < cov. matrix
multicollineation
- 3. 因子过多或少都不行.
- 4. Non-normal $\leftarrow$ sensitivity
t 分布

LM5 Economics and Investment Method. 经济变动如何影响

经济增长对实际无风险收益的影响. 1

Inter-temporal rate of substitution. (m 跨期替代率)

$$m = \frac{u_1}{u_0} < 1. \text{ 预期经济好, } u_1 \downarrow \text{ 边际效用递减, } m \downarrow.$$

① 均衡状态下, bond price $P_0 = m$, 不等时市场自动调节

② $P_0 = \sum \frac{CF_t}{(1+r)^t}$ { ① 单期 Bond: $P_0 = \frac{CF_1}{1+r}$ 收益率 $r = \frac{1-P_0}{P_0} = \frac{1}{P_0} - 1 = \frac{1}{m} - 1$ } L.M.F.H 0 — 1 — 1
P₀ — P₁

经济好 $m \downarrow$ $\uparrow r$
经济差 $m \uparrow$ $\downarrow r$

$\uparrow r \rightarrow m \downarrow, u_1 < u_0$, 现在消费效用更高

② 多期 Bond: $P_0 = \frac{ECP_1}{1+r} + \text{cov}(P_1, m)$ ① 通常, $P_1 \uparrow \rightarrow \text{wealth} \uparrow \rightarrow u_1 \downarrow$
 $m = \frac{u_1}{u_0}$ } $m \downarrow, \text{cov} < 0.$

② 经济极差 $\rightarrow$ 都买国债 $\rightarrow P_1 \uparrow$
 $\rightarrow u_1 \uparrow \rightarrow m \uparrow$ } $\text{cov} > 0.$

0 — 1 — 不确定 — S
 P₀ — ECP₁ — P₁

③ GDP Growth $\uparrow \rightarrow \text{risk} \uparrow \rightarrow P_0 \downarrow$, 说明分母折现率上升 $\rightarrow \uparrow r$

L5 GDPg, GDPg 正相关

经济对名义利率 Nominal Interest Rate 的影响. θ, π

$$\text{Nominal } R_f = 1 + \theta + \pi$$

预期内 Inflation, short term

不确定性带来的溢价 long-term

1. ST: Short-term 时: $1 + \theta$

Taylor Rule

Policy Rate $r = r^* + 0.5(i - i^*) + 0.5(Y - Y^*)$

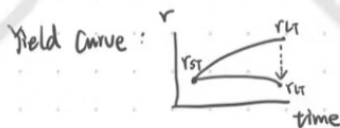
通胀 GDPg.

抗通胀与促增长的重要系数

① $i > i^* \rightarrow$ 过热 抗通胀
 $\uparrow r > r^* \rightarrow$ 加息 $\rightarrow$ 紧缩

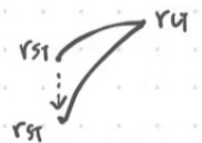
② $Y < Y^* \rightarrow$ 过冷 促增长
 $\downarrow r < r^* \rightarrow$ 降息 $\rightarrow$ 宽松

2. LT: Breakeven Inflation Rate (BEI): $BEI = (\theta + \pi) / (1 + \pi)$



If inverted, $r_{LT} < r_{ST}$, 预期经济衰退

During recession, policy rate 将被降低. yield curve steepen



Credit Premium 子 违约风险溢价

① $\delta = r_{\text{公司}} - r_{\text{相同期限国债}}$

② { 经济好 $\rightarrow \delta \downarrow, \text{risk} \downarrow$
经济差 $\rightarrow \delta \uparrow, \text{risk} \uparrow$

③ Expected Loss = Probability of Default $\times (1 - \text{Recovery Rate})$.

有抵押品. 经济好, 抵押品好卖 } Recovery Rate 高.

④ δ sensitivity. 即 δ 变化幅度

Factors { ① different ratings 信用评级 评级低, sensitivity 大.

② industrial sectors cyclical sector 大

③ company-specific 盈利状况差, sensitivity 大

⑤ Sovereign credit risk, 尤其 developing or emerging

经济影响股票市场. (股票市场额外溢价).

$$r_e = l + \theta + \pi + \gamma + k$$

股市风险溢价 $k = r_e - r_{\text{国债}}$

① $\gamma \uparrow \rightarrow k \uparrow$ 股票与信用风险正相关

② Bad Consumption Hedge 消费对冲: 经济差时, 价格下降. (除黄金外都是).

③ 经济好时, 多投 Growth Stocks, Small cap Stocks

经济对商业地产的影响.

Real Estate $\left\{ \begin{array}{l} \text{Rental Income 稳定 bond-like} \\ \text{Capital Gain 不稳定 stock-like} \end{array} \right.$

$$r = l + \theta + \pi + \gamma + k + \phi \quad \text{流动性溢价 } \phi$$

① 周期性特征, 周期敏感程度高. Bad Consumption Hedge

Summary

$R_i = l$	$\left\{ \begin{array}{l} \text{Nominal } R_f \end{array} \right.$	(长期国债 r)
$+ \theta$		
$+ \pi$		
$+ \gamma$	$\left\{ \begin{array}{l} \gamma \text{ 敏感性} \end{array} \right.$	(公司 r)
$+ k$		
$+ \phi$		
		(r_e)
		房地产
		Bad Consumption Hedge
		经济差 $\rightarrow P \downarrow \rightarrow R_p \uparrow$

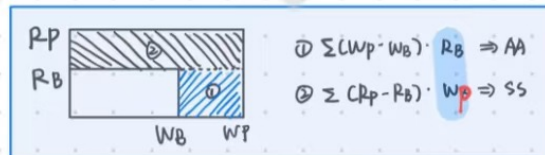
LN6. Active Portfolio Management.

RA 辨析

- Value Added (Active Return): $RA = R_p - R_B$
- Risk-Adjusted Value Added: $\alpha_A = R_p - \beta_p \cdot R_B$ 调整了风险.
- Active Weight. 积极权重: $\Delta W_i = W_p^i - W_B^i > 0$, 多配了好股.
- RA 贡献: $\Delta W_i \cdot RA^i$

RA Decomposition (不调整风险)

$RA \left\{ \begin{array}{l} \text{Asset Allocation } \Delta W \\ \text{Security Selection } \Delta R \end{array} \right.$



$$\Rightarrow RA = AA + SS$$

构建最优组合: 2 结论 + 3 公式

1. Sharpe Ratio

$$SR_p = \frac{R_p - R_f}{\sigma_p}$$

Sharp Ratio 不受现金 (无风险资产) 多少的影响。

2. Information Ratio

$$IR = \frac{RA}{GA} = \frac{R_p - R_B}{\sigma(R_p - R_B)}$$

IR 不受 Benchmark 比重影响。

no constraint 时, W 变化 无影响

3. Constructing Optimal Portfolio.

- $SR_{max} \Rightarrow SR_p^2 = SR_B^2 + IR^2$ $\therefore P$ 业绩 $\left\{ \begin{array}{l} \text{passive: } SR_B^2 \\ \text{active: } IR^2 \end{array} \right.$ * 是新组合!
- $GA^* = W_p \cdot GA \Rightarrow W_p = \frac{GA^*}{GA}$ $P^* \left\{ \begin{array}{l} \text{Fund } W_p \\ \text{Benchmark } 1 - W_p \end{array} \right.$
- $GA^* = \frac{IR \cdot \sigma_B}{SR_B} \cdot \sigma_B$

一般 GA 给定。
计算 GA^* 代入 W_p 从而构建
最优组合。 $SR_{p, max}^2 = SR_B^2 + IR^2$
passive active

Fundamental Law

$IC = \rho\left(\frac{w_i}{G_i}, \frac{RA_i}{G_i}\right)$ 描述预测与超额收益相关性 $\in [-1, 1]$.

$TC = \rho\left(\frac{w_i}{G_i}, \Delta w_i \cdot G_i\right)$ 描述落地执行能力。好票多配, 反应超额收益与配比如关系。若市场限制↑, 则 $TC \downarrow$ $\in [-1, 1]$

$BR = \frac{N}{1 + (N-1)\rho}$ 描述独立预测次数, N 是次数, ρ 是每次预测相关性。

1. Basic Fundamental Law 完美条件, 无 constraint. $\Rightarrow$ 默认 $TC=1$

$$IR^* = IC \cdot \sqrt{BR} \Rightarrow E(R_A)^* = IC \cdot \sqrt{BR} \cdot GA$$

2. Full Fundamental Law $TC < 1$

$$IR = TC \cdot IC \cdot \sqrt{BR} = TC \cdot IR_{Basic} \Rightarrow E(R_A) = TC \cdot TC \cdot \sqrt{BR} \cdot GA$$

预期 $RA \left\{ \begin{array}{l} \text{Ex Ante 事前: } E(R_A) = IR \cdot GA = TC \cdot IC \cdot \sqrt{BR} \cdot GA \\ \text{Ex Post 事后: } E(R_A | IC_{realized}) = TC \cdot IC_{realized} \cdot \sqrt{BR} \cdot GA \end{array} \right.$

真实 RA : $RA = \underbrace{E(R_A | IC_A)}_{\text{占 } TC^2} + \varepsilon$ noise 占 $1 - TC^2$

LM7 Trading Cost & Electronic Market.

Cost of Trading

1. 基础概念.

① 两大分类.

① Fixed Trading Cost {
Employing buy-side traders
Cost of equipping
Cost of office space

② Variable Transaction Cost {
Explicit cost
Implicit cost: 4类

Bid-Ask Spread: 做市商利润, 投资者成本

Market Impact 价格冲击: 大量买单推升价格. → 越买越贵.

Delay Cost 滑点成本: 决策与下单时间差导致的亏损, 易出现在外汇交易中.

Opportunity Cost: 放弃的收益, 交易中 Cancel 导致的亏损.

② Bid-Ask Spread & Order Books

Market bid-ask spread (Inside Spread) = $\min$ Best Ask - $\max$ Best Bid

Midquote price = (Best Ask + Best Bid) / 2.

★ 2. Costs of Trading

$$T.C = \text{side} \times (P - B) \times \text{size}.$$

{ 买 +1 卖 -1 }
↑ 真实基准 股数.

① Effective Spreads 有效价差.

利用下单时的 midquote price 作 benchmark. spread 一定是双边, round trip, 有买有卖

1. 计算: $ES = 2 \times \text{side} \times (P - \frac{Bid + Ask}{2}) \times \text{size}.$

2. {
ES > Market Spread: no price improvement.
ES = Market Spread
ES < Market Spread: price improvement. 比观测到最优的 spread 成本还低

3. 缺点: 只适用于 single trade < VWAP

不考虑 market impact

忽视 delay cost & opportunity cost → Implementation Shortfall

用后2种方法来修正.

② VWAP

1. 计算: $VWAP = \text{sum of the total dollar value of traders} / \text{total quantity of traders}$

$VWAP TC = \text{side} \times (Trade VWAP - VWAP benchmark) \times \text{size}.$

不×2!

2. 缺点: 大单交易 (Trader & market) or Same Rate 时, 都导致 TC 接近 0, 多适用小单.

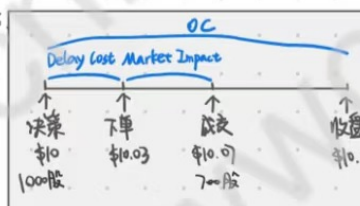
只看成交价格, 未考虑下单、撤单价.

③ Implementation Shortfall 执行落差.

1. 计算: $IS = \text{Paper Return} - \text{Actual Return}$ (理论收益 - 实际收益)

2. 优点: 完整考虑各成本

缺点: 计算复杂.



① Delay cost = $(10.03 - 10) \times 700$ (成本)

② Market Impact = $(10.07 - 10.03) \times 700$ (成本)

③ OC (撤单) = $(10.08 - 10) \times 700$ (成本)

3 部分隐性成本

> Execution Cost.

小红书

小红书号: 186439052

Advantages of Electronic Trading Systems.

- ① Cost cheap, narrow spread.
- ② Accuracy
- ③ Audit Trails 审计追踪.
- ④ Fraud Prevention: Hidden Order 更隐蔽.
- ⑤ Continuous Market. Event Risk ↓.
- ⑥ Efficiency of Computer.

★ Market Fragmentation 市场分割.

可以套利, ex: A股H股套利

可以利用算法整合流动性. { Super books
Smart order-routing (SOR) 智能最优路径.

Major Types of Electronic Traders.

- ① Electronic news-trader: information-motivated { 利好 → 买
利空 → 卖
- ② Electronic dealer: news trader 的对手方.
- ③ Electronic arbitrageurs
- ④ Electronic front runners: 算法观察别人订单, 抢先买断流动性. low-latency traders.
- ⑤ Electronic quote matchers: 匹配报价. 利用未成交的 standing order 套利.

★ Low Latency 低延时.

3方面: Taking (决策), Making (下单), Canceling (撤单) 都要快.

Reduce Latency: { Fast Communications 传输速度
Fast Computation 计算速度

Impact of Electronic Trading.

Advanced Orders

Trading Tactic: Immediate or Cancel Order (IOC) 帮助找到 hidden order.

Algorithms: 利用算法完成交易.

其他策略 { Hidden Order
Leapfrog 跳羊: Dealer 为吸引投资者, 不断降低 Ask Price → Spread ↓
Fllickering quotes 闪烁订单: 市场订单一会儿有一会儿无, 为防止 Leapfrog
Electronic Arbitrage.
Machine Learning.

Risk of Electronic Trading.

High-Frequency Traders (HFT) arms race → Unfair

System Risks { Runaway algorithms 程序错误
Fast finger errors 人为
Overlarge orders.
Malevolent order streams 恶意订单

Real-Time Surveillance 实时监控.

利用内幕消息做 Front Runner

账户间倒卖.

小单大买单.

不识别下单

市场操纵: Trade for market impact / Rumormongering / Wash Trading / Spoofing (Layering) / Bluffing /
Gunning the market 逼空 / Squeezing and Cornering 逼角